

Density of states and the steepest descent method: a non-iterative procedure

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A non-iterative approximation to the method of steepest descent is given for the evaluation of state density in a molecular system. Test calculations on 'large' and 'small' model molecules consisting of harmonic oscillators and rigid rotors show excellent agreement with the 'exact' first-order steepest-descent solutions, but are almost an order of magnitude faster in computing times.

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On présente une approximation non-itérative pour la méthode de descente la plus rapide pour l'évaluation de la densité d'états d'un système moléculaire. Des essais de calcul sur des molécules modèles "grande" et "petite" consistant d'oscillateurs harmoniques et de rotors rigides montrent un accord excellent avec les solutions "exactes" de descente la plus rapide du premier ordre, mais sont presque un ordre de grandeur plus rapide en temps de calcul.

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The density of states $N(E)$ of a model molecule at energy E is an important quantity in statistical rate theories (1, 2). Several approximate methods of calculation have been proposed in recent years, and extensive comparisons between them have been made by Forst and Prášil (3). Among various 'quantal' methods, the method of steepest descent (3-5) proves most attractive since it yields excellent approximations to the exact count, whilst requiring less computation than the method using Cauchy's residue theorem (6, 7).

The method of solution of the (first-order) steepest-descent approximation (4) has been derived independently by Forst and Prášil (3) and by Hoare and Ruijgrok (5): $N(E)$, formally the inverse Laplace Transform of the partition function, is related analytically to the saddle point of the integrand of the inversion integral; in this procedure, several iterations (10 to 15 for the first energy and typically 5 thereafter) are required to determine the saddle point. In the present paper, we report an alternative procedure which does not require such an accurate determination of the saddle point: a sufficiently accurate value for the saddle point can be constructed without iteration, with a consequent saving in computing time of a factor of between 5 and 10 in the calculation of $N(E)$. Since in unimolecular rate calculations $N(E)$ is required at some hundreds of values of the energy, the time saving can become appreciable.

The density of states $N_{vr}(E)$ for a system with v

independent vibrators and r free rotors¹ is formally the inverse Laplace Transform of the partition function $Z(\beta)$ (3-5), *i.e.*

$$[1] \quad N_{vr}(E) = \mathcal{L}^{-1}\{Z(\beta)\} = (2\pi i)^{-1} \times \int_{c-i\infty}^{c+i\infty} Z(\beta) \exp(\beta E) d\beta \equiv I$$

where $\beta = 1/kT$ is the transform parameter; the sum of states $G_{vr}(E)$ is readily related to $N_{v,r+2}(E)$ (1).

To evaluate I , one defines

$$[2] \quad \phi(\beta) = \ln [Z(\beta)] + \beta E$$

and the first-order steepest-descent method gives (3)

$$[3] \quad I \simeq \exp [\phi(\beta^*)][2\pi\phi''(\beta^*)]^{-1/2}$$

where β^* is the *saddle point*. Since

$$[4] \quad \phi'(\beta^*) = 0$$

$$[4a] \quad \left. \frac{-\partial \ln Z(\beta)}{\partial \beta} \right|_{\beta^*} = E$$

identifying β^* as the (inverse) temperature at which the average energy of the system is E (8). In the Forst-Prášil formulation (3), eq. 4 is transformed to an algebraic equation via $\theta = \exp(-\beta)$. The transformation, although mathematically convenient, makes subsequent

¹We follow the definition of Forst and Prášil (1, 3): a two-dimensional rotor is counted as two one-dimensional rotors.

computations more difficult. This is because, whereas β^* is between 0 and 10^{-3} , θ^* is between 0.999 and unity; thus, whilst a precision of 10^{-4} suffices for β^* , a precision of 10^{-10} is needed for θ^* (3). And it is precisely this less stringent requirement on the precision in β^* , together with the thermodynamic relation [4a], that permits a 'one-shot' approximation which yields accurate $N_{vr}(E)$. Note that using eq. 4a, one obtains

$$[5] \quad I \simeq Z(\beta^*) \exp(\beta^* E) \left[-2\pi \frac{\partial E(\beta)}{\partial \beta} \bigg|_{\beta^*} \right]^{-1/2}$$

The 'one-shot' approximation is achieved by relating β^* explicitly to β_0 , a first approximation, via the Taylor expansion of E , using [4a], viz.

$$[6a] \quad E(\beta^*) = E(\beta_0) + \Delta\beta \frac{\partial E}{\partial \beta} + \frac{1}{2}(\Delta\beta)^2 \frac{\partial^2 E}{\partial \beta^2} + O[(\Delta\beta)^3]$$

$$[6b] \quad = \frac{-\partial \ln Z}{\partial \beta} - \Delta\beta \frac{\partial^2 \ln Z}{\partial \beta^2} - \frac{1}{2}(\Delta\beta)^2 \frac{\partial^3 \ln Z}{\partial \beta^3} - O[(\Delta\beta)^3]$$

where $\Delta\beta = \beta^* - \beta_0$ and all derivatives are evaluated at β_0 . Now substituting the first-order approximation

$$[7] \quad \Delta\beta \simeq \Delta E \left(\frac{\partial E}{\partial \beta} \right)^{-1}$$

where $\Delta E = E(\beta^*) - E(\beta_0)$, back into the $(\Delta\beta)^2$ term of eq. 6a, one obtains

$$[8] \quad \beta^* \simeq \beta_0 + \Delta E \left(\frac{\partial E}{\partial \beta} \right)^{-1} \times \left[1 - \frac{1}{2} \Delta E \left(\frac{\partial E}{\partial \beta} \right)^{-2} \left(\frac{\partial^2 E}{\partial \beta^2} \right) \right]$$

$$[8a] \quad \equiv \beta_0 - \Delta E \left(\frac{\partial^2 \ln Z}{\partial \beta^2} \right)^{-1} \times \left[1 + \frac{1}{2} \Delta E \left(\frac{\partial^2 \ln Z}{\partial \beta^2} \right)^{-2} \left(\frac{\partial^3 \ln Z}{\partial \beta^3} \right) \right]$$

It might be thought that values of I thus computed might be grossly in error since [8] is only a second-order approximation to β^* , and the truncation error in β^* would be magnified by the exponential in eq. 5. However, since $\phi'(\beta^*) = 0$, a simple error analysis gives

$$[9] \quad \delta[\exp(\phi)] \simeq \exp(\phi) \phi' \delta\beta + O(\delta\beta)^2 \simeq O(\delta\beta)^2$$

which is necessarily small near β^* . This insensitivity arises because $\exp(\beta E)$ increases with β whereas $Z(\beta)$ decreases. In test calculations, I computed using [5] and [8] agrees with the exact steepest-descent solution to better than 1% even when β_0 differs from β^* by as much as 20%, except for very large r at low E (since then $Z(\beta) \propto \beta^{-r/2}$). Thus, the ultimate performance of this approximation hinges upon the initial choice of β_0 . In practice, however, where $N_{vr}(E)$ is computed at hundreds of energies, typically only 50–100 cm^{-1} apart, β^* for two neighbouring energies are not very different. Thus, except for the first cycle, we may take β_0 to be the β^* value corresponding to the preceding energy, *fully exploiting the fact that in* [8], β_0 , ΔE and $(\partial E/\partial \beta)|_{\beta_0}$ are all known quantities: β^* is readily determined by explicit evaluation of $(\partial^2 E/\partial \beta^2)|_{\beta_0}$; no iteration is needed. This leaves only the choice of the initial value of β_0 at the starting energy, and for r rigid rotors and n groups of harmonic oscillators of degeneracies g_i and frequencies ν_i ($\nu_1 < \nu_2 < \dots < \nu_n$ and $\sum_i^n g_i = v$), the following empirical relation is convenient²

$$[10] \quad \beta_0 = \frac{r}{2E} + \frac{m}{S_m} \ln \left(1 + \frac{S_m}{E} \right)$$

where

$$S_m = \sum_{i=1}^m g_i \nu_i$$

with

$$m = 1 \quad \text{if} \quad g_1 \nu_1 \geq E$$

or

$$m = n \quad \text{if} \quad S_n \leq E$$

and

$$S_m \leq E < S_{m+1}$$

otherwise.

To test the performance of the method, we have computed $N_{vr}(E)$ for two model molecules for which both exact direct counts and exact steepest-descent solutions are available (3): they are model *A*, representing a typical 'small' molecule and model *B* representing a typical 'large' molecule, both with varying numbers of rotors. Tables 1 and 2 compare these previous calculations with those using eqs. 10, 8, and 5;

²Because of the insensitivity of I to β_0 (see text above), any empirical relation which gives a good approximation to β^* will suffice: the choice of [10] is based on the realisation that β_0 is exactly β^* for a system of r rotors or v degenerate oscillators.

TABLE 1. Reduced density $N_{vr}'(E)$ for model A*

r	E (cm^{-1})	Exact count (ref. 3)	Steepest descent	
			Exact (ref. 3)	This work
2	2 000	5	5.41	5.42
	5 000	35	37.70	37.72
	10 000	291	298.0	298.2
	20 000	3730	3801	3800
4	2 000	4624	4703	4717
	5 000	5.500×10^4	5.725×10^4	5.739×10^4
	10 000	7.102×10^5	7.266×10^5	7.279×10^5
	20 000	1.557×10^7	1.582×10^7	1.583×10^7

The frequency pattern is (3): 880, 1296, 1440, 3774, 3788 cm^{-1} .TABLE 2. Reduced density $N_{vr}'(E)$ for model B

r	E (cm^{-1})	Exact count (ref. 3)	Steepest descent	
			Exact (ref. 3)	This work
2	2 000	2.0×10^1	1.616×10^1	1.612×10^1
	5 000	6.2×10^2	5.747×10^2	5.750×10^2
	10 000	3.677×10^4	3.729×10^4	3.722×10^4
	20 000	1.052×10^7	1.066×10^7	1.067×10^7
4	2 000	8.753×10^3	9.576×10^3	9.598×10^3
	5 000	5.009×10^5	5.180×10^5	5.182×10^5
	10 000	4.783×10^7	4.869×10^7	4.867×10^7
	20 000	2.124×10^{10}	2.145×10^{10}	2.147×10^{10}
12	2 000	3.052×10^{14}	3.144×10^{14}	3.252×10^{14}
	5 000	1.383×10^{17}	1.409×10^{17}	1.445×10^{17}
	10 000	6.815×10^{19}	6.917×10^{19}	6.930×10^{19}
	20 000	1.855×10^{23}	1.873×10^{23}	1.879×10^{23}

*The frequency pattern is (3): 983(4), 1415(4), 2000(1), 3034(6) cm^{-1} .

following Forst and Prášil (3), the tabulated quantities are *reduced* state densities, *i.e.*

$$[11] \quad N_{vr}'(E) = N_{vr}(E)/Q_r'$$

where Q_r' is the classical partition function for r rotors with the factor $(kT)^{r/2}$ deleted. Note that in practice, one would not use eq. 10 to estimate β_0 at each value of the energy, as we have done in this test, but one would use the value of β^* calculated at the preceding value of the energy; this will be a better approximation than [10], so that the method will actually perform better than is shown in Tables 1 and 2 in any practical application.

We conclude by noting that since the desired accuracy of $N_{vr}(E)$ in unimolecular rate calcula-

tions is of the order of a few percent, this method should prove efficient and useful in the entire range where the standard steepest-descent method is applicable: at low energy, direct counts, for which efficient algorithms exist (9, 10), may be used. We note also that the accuracy of the approximation improves as E increases, making it attractive for calculations in the very high energy range, such as are needed in mass-spectral applications.³ Finally, generalisation to more

³An annotated FORTRAN program listing (including test cases and specimen output) is available upon request, at a nominal charge, from The Depository of Unpublished Data, CISTI, National Research Council of Canada, Ottawa, Canada K1A 0S2.

realistic systems, *e.g.* Morse oscillators, is straightforward.

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